Digital Material Passport

ID **DMP-METAL-007** - *Version* **1.0.0**
Issue Date **2025-05-20** - *Certificate Type* **EN 10204 3.1**

Manufacturer	Customer	Business Transaction
ACME Metal Works GmbH Industrial Park 123 52066 Aachen DE quality@acme-metal.example.com	**Bridge Constructors Europe Ltd.** Engineering Plaza 456 75001 Paris FR procurement@bridgeconstructors.example.com	**Order** PO-98765 · **20000 kg** Pos. 5 · 2025-04-20 **Delivery** DN-12345 · **20000 kg** Pos. 1 · 2025-05-19

Certificate Receiver
Civil Infrastructure Authority
Government Building 789
Administration Wing
1000 Brussels
BE
inspections@infra-authority.example.gov.be

Product

Product Name	Structural Steel S420N Plate	*Name (EN)*	1.8902
Batch ID	H-45678-01	*Shape*	Plate - Width 2500 mm - Thickness 25 mm - Length 12000 mm
Surface Condition	Shot blasted and primed	*Product Norm*	EN 10025-3 (2019)
Weight	20000 kg	*Product Norm*	EN 1090-2 (2018)
Production Date	2025-05-18		
Country of Origin	DE		

Chemical Analysis

Heat Number **H-45678** - *Melting Process* **EAF+LF+VD** - *Casting Method* **ContinuousCasting** - *Casting Date* **2025-05-17** - *Sample Location* **Ladle**

Symbol	*Unit*	*Min*	*Max*	*Actual*
C	%		0.22	0.16
Mn	%	1	1.7	1.38
Si	%		0.6	0.32
P	%		0.025	0.016
S	%		0.02	0.008

Symbol	*Unit*	*Min*	*Max*	*Actual*
Nb	%		0.05	0.022
V	%		0.05	0.034
Ti	%		0.03	0.009
CEV	%		0.48	0.4

CEV = C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15: 0.4%

Mechanical Properties

Property	*Symbol*	*Actual*	*Minimum*	*Maximum*	*Method*	*Status*
Tensile Strength[1]	Rm	548 / 550 / 552 MPa - Average 550 - Min 548 - Max 552	520 MPa	680 MPa	EN ISO 6892-1	✓
Yield Strength[1]	ReH	438 / 440 / 442 MPa - Average 440 - Min 438 - Max 442	420 MPa		EN ISO 6892-1	✓
Elongation after fracture 3 specimens tested, gauge length 5.65#S_0, transverse to rolling direction	A	20.5 / 21 / 21.5 % - Average 21 - Min 20.5 - Max 21.5	19 %		EN ISO 6892-1	✓
Charpy V-notch Impact Energy 3 specimens tested at -20°C, transverse to rolling direction	KV	63 / 65 / 67 J - Average 65 - Min 63 - Max 67 - Std Dev 2	40 J		EN ISO 148-1	✓

[1] 3 specimens tested at 20°C, transverse to rolling direction

Supplementary Tests

Test	*Result / limits*	*Method*	*Status*
Ultrasonic Testing Class S2E2	Yes - No recordable indications exceeding acceptance criteria	EN 10160	✓
Through-thickness Properties	Z25	EN 10164	✓
Weldability	Yes - Satisfactory welding properties	Internal Method based on EN ISO - 15614-1	✓

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with the requirements of EN 10025-3:2019 and EN 10204:2004 type 3.1. The product complies with the Construction Products Regulation (EU) No 305/2011 and is suitable for use in structural applications according to EN 1090-2:2018, up to and including Execution Class EXC3.

Validated by **John Smith**, Quality Manager, Quality Assurance - 2025-05-20

Validated by **Maria Schmidt**, Quality Manager, Quality Assurance - 2025-05-20